

Diego Chuquillanqui Cuenca

During the first two weeks of the Sensing the World module, I was introduced to materials science, computer engineering, circuits, and sensor fabrication through several hands-on activities. In the first class, we discussed what materials scientists and computer engineers work with, what they create, and the types of problems they solve, and completed a diagnostic activity to see our starting familiarity with concepts related to the module. In the second class, we visited the maker space labs and learned about resources available to students, including 3D printers, laser cutters, and CNC machines. After the tour, I worked with my assigned group of three to measure the resistance of copper using a multimeter. We then measured the dimensions of the copper and used those measurements, along with its resistance, to calculate its resistivity. This also helped show why copper is commonly used as a conductor because of its low resistivity.

In the third class, we visited the Sinclair cleanroom and learned about processes such as MOCVD and photolithography, which allow very thin layers of materials to be grown and patterned precisely. After the tour, my group and I measured the resistivity of Ag paste, Ag paint, and carbon paint, with each person focusing on one material. I worked with Ag paint by using tape as a mask on a glass slide, cutting out a strip, applying the paint, removing the excess with another glass slide, and drying the finished strip on a hot plate. I then measured its resistance and dimensions and used them to calculate its resistivity, following the same basic process we had used with copper. In the fourth class, I made a butterfly-shaped paper circuit using copper tape, LEDs, a battery, and a paper switch. Folding the paper completed the circuit and caused the LEDs to light up. Afterward, I made interdigitated electrodes with Ag paint on glass and spin-coated PEDOT:PSS over them. Once the material dried, I measured its resistance before and after breathing on it and observed a change when it was exposed to water vapor.

These activities helped me understand how the properties of materials can be used to turn a physical input into an electrical output. Different materials have different properties, so choosing the right material is an important part of designing a sensor. For example, the PEDOT:PSS responded to the water vapor in my breath by changing its resistance, allowing humidity to produce a measurable electrical change. I also better understood why interdigitated electrodes are useful. Their alternating fingers create many small gaps between the two electrodes, giving the sensing material more places to connect them and making changes in its electrical behavior easier to detect. The two tours also showed me different sides of engineering fabrication: the maker space focused on tools that can be used to quickly build and prototype designs, while the cleanroom showed how much more precise processes are used to manufacture very small electronic devices.

Making the Ag paint strip also showed me how small changes in the fabrication process can affect the result. On my first attempt, I heated the sample before removing the tape, and part of the Ag paint was scraped off when I later removed it. I had to make the sample again, but this time I removed the tape before heating it. This made me realize that in fabrication, the order of the steps can directly affect the quality of the final device. Working with the same group throughout these activities was also useful because we could compare our results and help each other when something did not work as expected. Overall, the hands-on activities helped me see how material selection, fabrication, and electrical measurements come together to create a working sensor.